

H524 Introduction to Biostatistics

Mid-Semester Individual Assignment

AI-Assisted Statistical Analysis

Fall 2025

Dr. John Molitor

Due: Week 6 (Friday by 11:59 PM)

Assignment Overview

This individual assignment replaces the traditional midterm exam and serves as preparation for your group project. You will conduct a focused statistical analysis using AI tools, demonstrating your understanding of descriptive statistics, hypothesis testing, and AI integration principles covered in the first half of the course.

1 Learning Objectives

By completing this assignment, you will demonstrate your ability to:

- Apply descriptive statistical methods to real health data
- Conduct appropriate hypothesis tests and interpret results
- Effectively use AI tools for statistical analysis while maintaining scientific rigor
- Document and verify AI-generated outputs
- Communicate statistical findings clearly and accurately

2 Assignment Components

2.1 Dataset Selection

Choose **ONE** of the following simplified datasets:

2.1.1 Option 1: Heart Disease Risk Factors

- **Data:** Framingham Heart Study subset (provided on Canvas)
- **Variables:** Age, gender, BMI, systolic BP, cholesterol, smoking status, diabetes status
- **Sample Size:** 500 participants
- **Focus:** Cardiovascular risk factor analysis

2.1.2 Option 2: COVID-19 Vaccination Outcomes

- **Data:** Simulated vaccination study data (provided on Canvas)
- **Variables:** Age group, vaccination status, breakthrough infections, hospitalization
- **Sample Size:** 1000 participants
- **Focus:** Vaccine effectiveness analysis

2.1.3 Option 3: Mental Health and Exercise

- **Data:** Depression screening study (provided on Canvas)
- **Variables:** Age, gender, exercise frequency, depression score, anxiety score
- **Sample Size:** 400 participants
- **Focus:** Mental health and lifestyle factors

3 Required Analysis Tasks

3.1 Task 1: Descriptive Analysis (25 points)

1. Data Exploration:

- Import and examine the dataset structure
- Check for missing values and outliers
- Create summary statistics for all variables

2. Descriptive Statistics:

- Calculate measures of central tendency and variability for continuous variables
- Create frequency tables for categorical variables
- Generate appropriate summary tables

3. Data Visualization:

- Create 3-4 appropriate graphs (histograms, box plots, bar charts)
- Include proper titles, labels, and legends
- Briefly interpret each visualization

3.2 Task 2: Hypothesis Testing (30 points)

1. Research Question:

- Formulate one clear, testable research question based on your chosen dataset
- State your null and alternative hypotheses

2. Statistical Test Selection:

- Choose the appropriate statistical test (t-test, chi-square, ANOVA)

- Justify your choice based on variable types and research question
- Check test assumptions

3. Analysis and Interpretation:

- Conduct the statistical test
- Report test statistic, p-value, and confidence interval (if appropriate)
- Interpret results in the context of your research question
- Discuss practical significance vs. statistical significance

3.3 Task 3: AI Integration and Documentation (25 points)

1. AI Tool Usage:

- Use GitHub Copilot for at least 50% of your R code generation
- Use an AI assistant (ChatGPT, Claude, etc.) for interpretation guidance
- Document each instance of AI use with the provided template

2. Verification Process:

- Manually check key statistical calculations
- Verify AI-generated interpretations against course materials
- Document any corrections made to AI outputs

3. AI Usage Log:

- Complete the provided AI documentation template
- Include specific prompts used and outputs received
- Reflect on benefits and limitations of AI assistance

3.4 Task 4: Written Report (20 points)

1. Report Structure (3-4 pages):

- Introduction with research question and dataset description
- Methods section describing your analytical approach
- Results section with tables, figures, and statistical findings
- Brief discussion of findings and limitations
- References (course materials and dataset sources)

2. Communication Quality:

- Clear, professional writing appropriate for a public health audience
- Proper statistical terminology and interpretation
- Well-formatted tables and figures with informative captions

4 Deliverables

Submit the following files via Canvas by the deadline:

1. **R Script File (.R):**

- Well-commented R code for all analyses
- Clear section headers for each task
- Working code that reproduces all results

2. **Written Report (.pdf):**

- 3-4 page report following the specified structure
- Include all tables and figures within the report
- Professional formatting and presentation

3. **AI Usage Documentation (.pdf):**

- Completed AI documentation template
- Detailed log of all AI interactions
- Reflection on AI assistance experience

4. **Data File (.csv):**

- Your cleaned dataset (if modifications were made)
- Include variable labels and coding information

5 AI Usage Requirements

Required AI Integration:

1. **GitHub Copilot:** Use for R code generation, debugging, and optimization
2. **AI Assistant:** Use for statistical interpretation and method selection guidance
3. **Documentation:** Record all AI interactions using the provided template
4. **Verification:** Independently verify all AI-generated content

AI Documentation Template:

```
## AI Usage Entry [Number]
Date: [Date]
Tool: [GitHub Copilot/ChatGPT/Claude/etc.]
Task: [Specific statistical task]
Prompt: [Exact prompt or request]
AI Output: [Summary of response]
Verification Method: [How you checked the output]
Final Decision: [What you used/modified]
Reflection: [Benefits and limitations observed]
```

6 Grading Rubric

Component	Points	Criteria
Descriptive Analysis	25	Comprehensive data exploration and visualization
Hypothesis Testing	30	Appropriate test selection and correct interpretation
AI Integration & Documentation	25	Effective AI use with thorough documentation
Written Report	20	Clear communication and professional presentation
Total	100	

6.1 Detailed Grading Criteria

6.1.1 Excellent (90-100%)

- Comprehensive and accurate statistical analysis
- Effective AI integration with complete documentation
- Clear, professional writing with correct interpretations
- Well-designed visualizations and properly formatted tables

6.1.2 Good (80-89%)

- Mostly correct analysis with minor computational errors
- Good AI usage with adequate documentation
- Generally clear writing with mostly correct interpretations
- Appropriate visualizations with minor formatting issues

6.1.3 Satisfactory (70-79%)

- Basic analysis completed with some errors
- Limited AI integration or incomplete documentation
- Unclear writing or some misinterpretations
- Basic visualizations that convey main findings

6.1.4 Needs Improvement (<70%)

- Major errors in statistical analysis or interpretation
- Minimal AI usage or poor documentation
- Poor communication or significant misunderstandings
- Inadequate or missing visualizations

7 Resources and Support

7.1 Statistical Resources

- Course lecture notes and textbook (Pagano & Gauvreau)
- R documentation and help files
- Canvas discussion board for questions
- Office hours: Wednesday 10:00-11:30 AM

7.2 AI Tool Access

- GitHub Copilot (required - set up in RStudio)
- ChatGPT, Claude, or similar AI assistant for consultation
- Course-provided AI prompt templates

7.3 Technical Requirements

- R and RStudio (latest versions)
- Required R packages: tidyverse, ggplot2, stats
- PDF generation capability for reports

8 Academic Integrity

Important Reminders:

- This is an **individual assignment** - no collaboration with classmates
- All AI-generated content must be verified and properly documented
- You are responsible for the accuracy of all submitted work
- Failure to document AI usage constitutes academic dishonesty
- Submitting unverified AI output as your own work is plagiarism

9 Tips for Success

1. **Start Early:** Data analysis often takes longer than expected
2. **Document Everything:** Keep detailed notes of your analysis decisions
3. **Verify AI Output:** Always double-check AI-generated code and interpretations
4. **Ask Questions:** Use office hours if you encounter difficulties
5. **Focus on Understanding:** Ensure you can explain every step of your analysis
6. **Practice Presentation:** You may be asked to briefly present your findings in class

10 Late Submission Policy

- Late submissions will be penalized 10% per day
- Extensions may be granted for documented emergencies
- Technical difficulties are not grounds for extensions - submit early to avoid issues